$M = 100$ puts, $P^{BS}(t,s)$ the put price, $\delta_t = \partial_s P^{BS}(t,S_t) =$

$= N(d_1) - 1 < 0$. Per the sheet, scaled by M: $Y_t = 0$, $h_t = M\delta_t$,

$Y_t^{new} = Y_t - h_t S_t = -h_t S_t$

$$V_t = Y_t^{new} + h_t S_t - M P^{BS}(t, S_t) = -M P^{BS}(t, S_t)$$

with $S_{t+\Delta} = S_t e^x$, $x = X_{t+\Delta} \sim N((\mu - \frac{1}{2}\sigma^2)\Delta, \sigma^2\Delta)$,

$$V_{t+\Delta} = h_t S_t e^x + Y_t^{new} e^{r\Delta} - M P^{BS}(t+\Delta, S_t e^x) =$$

$$= h_t S_t (e^x - e^{r\Delta}) - M P^{BS}(t+\Delta, S_t e^x)$$

$$L_{t+\Delta} = V_t - V_{t+\Delta} = M[P^{BS}(t+\Delta, S_t e^x) - P^{BS}(t, S_t)] -$$

$$- M\delta_t S_t (e^x - e^{r\Delta}).$$

where $P^{BS} = 27.11$, $\delta_t = -0.6086$, $h_t = -60.86$ shares

$V_t = -2,710.77$

Q 1

For a constant dollar position $\vartheta \in \mathbb{R}^d$ with linearized losses $L_{t+j\Delta} = -\vartheta^T X_{t+j\Delta}$ and $X_{t+j\Delta}$ i.i.d. $N(0, \Sigma)$

$$L^1_{t+J\Delta} = \sum_{j=1}^{J} L_{t+j\Delta} = -\vartheta^T \sum_{j=1}^{J} X_{t+j\Delta}, \quad \sum_{j=1}^{J} \sim N(0, J\Sigma),$$

$$L^t_{t+J\Delta} \sim N(0, J\vartheta^T\Sigma\vartheta)$$

so $\mathrm{VaR}_\alpha\left(L^{+}_{t+J\Delta}\right) = \sqrt{J\delta^\top \Sigma \delta}\, q_{p^{-1}}(\alpha) = \sqrt{J}\,\mathrm{VaR}_\alpha(L_{t+\Delta})$

The rule needs linear losses, zero mean, independence across days and normality

Q2.

Grid $T_j = t + j\Delta$, $j = 0, \dots, J$ Period 1 is the setup above with $T_0 = t$ At the start of period $j \geq 2$ the hedge is reset to $h_{T_{j-1}} = M\delta(T_{j-1}, S_{T_{j-1}})$ and the share trade is paid from cash:

$$Y^{new}_{T_{j-1}} = Y_{T_{j-1}} + (h_{T_{j-2}} - h_{T_{j-1}})\, S_{T_{j-1}}, \quad Y_{T_j} = Y^{new}_{T_{j-1}} e^{r\Delta}$$

$$V_{T_j} = Y_{T_j} + h_{T_{j-1}} S_{T_j} - M P^{BS}(T_j, S_{T_j}), \quad L_{T_j} = V_{T_{j-1}} - V_{T_j}$$

Rebalancing is self-financing: the value just after the trade $Y^{new}_{T_{j-1}} + h_{T_{j-1}} S_{T_{j-1}} - M P^{BS}$ equals the value just before, $Y_{T_{j-1}} + h_{T_{j-2}} S_{T_{j-1}} - M P^{BS}$, so $V_{T_{j-1}}$ is well defined. The daily losses therefore telescope

$$L^{t}_{t+J\Delta} = \sum_{j=1}^{J} (V_{T_{j-1}} - V_{T_j}) = V_{T_0} - V_{T_J}$$

Q 3

Expand the one day loss to second order about (t, S_t)

$$e^x - 1 = x + \frac{1}{2}x^3 + O(x^3), \quad e^x - e^{r\Delta} = x + \frac{1}{2}x^2 - r\Delta + \dots$$

$$L_{t+\Delta} = M\left[\theta\Delta + \delta S_t\left(x + \tfrac{1}{2}x^2\right) + \tfrac{1}{2}\gamma S_t^2 x^2\right] - M\delta S_t\left[x + \tfrac{1}{2}x^2 - r\Delta\right] + \dots =$$

$$= M\left[\left(\theta + rS_t\delta\right)\Delta + \tfrac{1}{2}\gamma S_t^2 x^2\right]$$

The term linear in x cancels. By the Black Scholes PDE

$$\theta + rS\delta + \tfrac{1}{2}\sigma^2 S^2\gamma = rP^{BS}$$

$$L_{t+\Delta} \approx M_r P^{BS}\Delta + \frac{1}{2}M\gamma S_t^2\left(x^2 - \sigma^2\Delta\right) \approx b\sigma^2\Delta\left(Z^2 - 1\right)$$

$$b = \tfrac{1}{2}M\gamma S_t^2, \quad Z \sim N(0,1)$$

Since $M_r P^{BS}\Delta = 0.01$ is negligible and $x \approx \sigma\sqrt{\Delta}\,Z$.

So assumption (i) fails: the hedged one day loss is not linear but quadratic in the risk factor. centered scaled χ_1^2 with mean ≈ 0 and skewness 2.83.

Daily rehedging resets the delta, so the ten-day loss is a sum of 10 independent copies, $L^t_{t+10\Delta} \approx b\sigma^2\Delta\,(Q - 10)$ with $Q \sim \chi^2_{10}$. Quantile scale with the multiplier so

$$\frac{VaR_{0.95}\left(L^{+}_{t+10\Delta}\right)}{\sqrt{10}\,VaR_{0.95}\left(L_{t+\Delta}\right)} \approx \frac{q_{0.95}\left(\chi^2_{10}\right)-10}{\sqrt{10}\left(q_{0.95}\left(\chi^2_{10}\right)-1\right)} = \frac{18.307-10}{3.162\times 2.841} = 0.925$$

against 0.94 from the simulation.

Assumption (iv) fails and rule overstates 10-day VaR by 6-8%.

The $\sqrt{10}$ rule scales the loss of freshly hedged book, so it implicitly assumes daily re-hedging and lands near that estimate, not frozen one.

Code: https://github.com/zhurleonid/corporate-risk-management
(`homework_3/part3_var_10day/hw3_part3_code.ipynb`)

Homework 3 comes as three problem sheets; this is sheet 3, numbered Q1 to Q3 as on the sheet. The write-up is below. The code is appended at the end. The derivations are handwritten and attached separately: the one-day loss of the hedged position (set-up), the square-root-of-time rule (Q1), the re-hedging recursion and its aggregation (Q2), and the reason the rule fails here (Q3). The write-up points to them where they belong.

A note on the problem sheet

The sheet says the hedge ratio is $h_t = \partial_s P^{BS}(t, S_t)$ "where P^{BS} denotes the Black-Scholes call price". The position is a short put, so I read P^{BS} as the put price. Its delta is $N(d_1) - 1 < 0$, so the hedge is a short stock position, about 61 shares for 100 puts.

10-day VaR of a delta-hedged short put

Set-up

Short 100 puts with $\kappa = 170$ and $T = 0.292$, hedged with the stock under

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t, \qquad X_{t+\Delta} = \ln \frac{S_{t+\Delta}}{S_t} \sim N\Big(\Big(\mu - \tfrac{1}{2}\sigma^2\Big)\,\Delta,\ \sigma^2\Delta\Big)$$

with $S_t = 152.51$, $r = 0.119\%$, $\sigma = 49.07\%$, $\mu = 16.91\%$, $\Delta = 1/252$ and $J = 10$.

> **By hand, attached:** the one-day loss of the hedged position (set-up), the square-root-of-time rule (Q1), the daily re-hedging recursion and its aggregation (Q2). The pseudocode below implements those expressions.

Time t quantities for 100 puts:

$$P^{BS}(t, S_t) = 27.1077, \qquad \delta = N(d_1) - 1 = -0.6086,$$

$$h_t = 100\,\delta = -60.86 \text{ shares}, \qquad V_t = -2{,}710.77.$$

The hedge is a short of 61 shares, and V_t is minus the premium of the puts.

Pseudocode (Q1 and Q2)

```
inputs: mu, sigma, r, T, Delta, S0, kappa, M = 100 puts,
alpha
        J = 10 days, N = 40,000 runs
```

```
m  = (mu - sigma^2/2)*Delta,   s = sigma*sqrt(Delta)
h0 = M * (N(d1(T, S0)) - 1),   V_t = -M * P_BS(T, S0)

Q1, one-day VaR and square-root-of-time:
    draw X_1 ... X_N iid N(m, s^2)
    for each i:
        S1    = S0 * exp(X_i)
        Y_new = 0 - h0*S0
        V1    = h0*S1 + Y_new*exp(r*Delta) - M*P_BS(T -
Delta, S1)
        L_i   = V_t - V1
    VaR_1 = sorted L at position ceil(N*alpha) = 38,000
    VaR_sqrt = sqrt(10) * VaR_1

Q2, 10-day VaR with daily re-hedging:
    draw an N x J matrix of X_{i,j} iid N(m, s^2)
    for each path i:
        h = h0,  Y_new = -h0*S0,  S_prev = S0,  V_prev = V_t
        for j = 1 ... J:
            S_j   = S_prev * exp(X_{i,j})
            Y     = Y_new * exp(r*Delta)
            V_j   = Y + h*S_j - M*P_BS(T - j*Delta, S_j)
            l_j   = V_prev - V_j                    loss over
day j
            h_new = M * (N(d1(T - j*Delta, S_j)) - 1)   re-
hedge
            Y_new = Y + (h - h_new)*S_j             cash pays
the trade
            h, S_prev, V_prev = h_new, S_j, V_j
        L_i = l_1 + ... + l_J                       = V_t -
V_J
    VaR_10 = sorted L at position 38,000

reference: same paths, hedge frozen at h0 for all ten days
```

The one-day and ten-day runs use separate draws from the same seeded generator, so the first day of the ten-day paths is not the one-day sample; the two one-day loss distributions agree to within Monte Carlo noise.

Q1 and Q2. Results

	$\mathrm{VaR}_{0.95}$	mean	s.d.	skewness
Q1: one-day loss	29.80	−0.02	14.86	2.83
Q1: $\sqrt{10} \times$ one-day VaR	94.24			
Q2: 10-day loss, re-hedged daily	88.62	0.11	48.00	0.89
reference: 10-day loss, hedge frozen at t	306.41	0.85	156.61	3.02

Monte Carlo with 40,000 runs, in dollars, 100 puts.

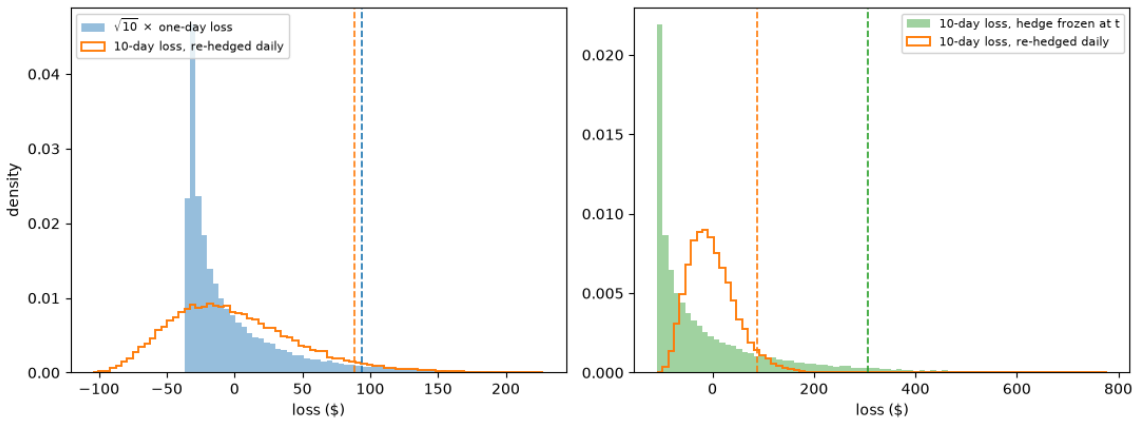
Q3. Comparison

> **Q3: the reason for the difference by hand, attached**, including the frozen-hedge contrast.

The square-root-of-time rule overstates the 10-day VaR by 6%. The time-aggregated estimate is 88.62 against 94.24 from scaling, a ratio of 0.940; the attached argument predicts 0.925. This is the same pattern as the lecture's hedged call, 87.66 against 90.96.

The simulated shape statistics in the table are what the argument relies on. From one day to ten the standard deviation scales with $\sqrt{10}$ to within 2%, while the skewness falls from 2.83 to 0.89 and the distribution loses its one-sided shape, as the left panel of the figure shows. Both means are within a dollar of zero, so the drift plays no part in the gap.

Re-hedging daily is what makes the 10-day number small at all. With the hedge frozen at t for ten days, which is the Homework 2 set-up, the $\mathrm{VaR}_{0.95}$ on the same paths is 306.41, three and a half times the re-hedged figure and close to the 304.52 found there; the attached argument predicts a ratio of 3.4. The right panel of the figure shows the frozen-hedge distribution as a spike at the theta floor with a tail reaching past 400, against the re-hedged distribution's near-bell shape.



Left: $\sqrt{10}$ times the one-day loss against the ten-day loss with daily re-hedging, with the two $\mathrm{VaR}_{0.95}$ estimates as dashed lines. Right: the ten-day loss with the hedge frozen at t against the re-hedged one, on the same 40,000 paths. The top 0.5% is cut off in both panels.

Appendix: source code

The code below produces every figure and number above.

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

ALPHA = 0.95
DELTA = 1 / 252

plt.rcParams['figure.figsize'] = (11, 5)


def var_emp(losses, a):
    # VaR from a sample: the ceil(n*a)-th order statistic (Lemma 7 in the nc
    ls = np.sort(np.asarray(losses))
    n = len(ls)
    return ls[int(np.ceil(n * a)) - 1]
```

In [2]:
```
mu_bs = 0.1691
sigma = 0.4907
r = 0.00119
T = 0.292
S0 = 152.51
K = 170.0
M_PUT = 100          # short 100 puts
J_DAYS = 10
N_RUNS = 40_000


def d1(tau, S):
    return (np.log(S / K) + (r + sigma**2 / 2) * tau) / (sigma * np.sqrt(tau

def d2(tau, S):
    return d1(tau, S) - sigma * np.sqrt(tau)

def put(tau, S):
    return K * np.exp(-r * tau) * norm.cdf(-d2(tau, S)) - S * norm.cdf(-d1(t

def put_delta(tau, S):
    return norm.cdf(d1(tau, S)) - 1


P0 = put(T, S0)
h0 = M_PUT * put_delta(T, S0)
V_t = -M_PUT * P0                    # Y_t = 0, so V_t = Y_new + h S - M P =
print('put price  %.4f   delta  %.6f   shares h_t = %.3f' % (P0, put_delta(T
print('V_t = %.2f' % V_t)
```

```
put price  27.1077   delta  -0.608555   shares h_t = -60.855
V_t = -2710.77
```

In [3]:
```
rng = np.random.default_rng(2026)
m_x = (mu_bs - sigma**2 / 2) * DELTA
s_x = sigma * np.sqrt(DELTA)

x1 = rng.normal(m_x, s_x, N_RUNS)
S1 = S0 * np.exp(x1)
```

```
Y_new = 0.0 - h0 * S0
V1 = h0 * S1 + Y_new * np.exp(r * DELTA) - M_PUT * put(T - DELTA, S1)
L1 = V_t - V1

var_1d = var_emp(L1, ALPHA)
var_sqrt = np.sqrt(J_DAYS) * var_1d
print('one-day loss: mean %.3f   sd %.3f   min %.3f' % (L1.mean(), L1.std(dd
print('one-day VaR_0.95        = %.4f' % var_1d)
print('sqrt(10) x one-day VaR  = %.4f' % var_sqrt)
```

```
one-day loss: mean -0.023   sd 14.862   min -10.582
one-day VaR_0.95        = 29.8005
sqrt(10) x one-day VaR  = 94.2376
```

In [4]:

```
rng = np.random.default_rng(2026)
x_path = rng.normal(m_x, s_x, (N_RUNS, J_DAYS))     # X_{t+j*Delta}, j = 1..
S_path = S0 * np.exp(np.cumsum(x_path, axis=1))
S_prev = np.full(N_RUNS, S0)

# period 1: hedge set at t, cash Y_t = 0
Y = np.full(N_RUNS, 0.0)
h = np.full(N_RUNS, h0)
Y_new = Y - h * S_prev
V_prev = Y_new + h * S_prev - M_PUT * put(T, S_prev)        # = V_t on every
daily_loss = np.empty((N_RUNS, J_DAYS))

for j in range(1, J_DAYS + 1):
    tau_j = T - j * DELTA
    S_j = S_path[:, j - 1]
    Y = Y_new * np.exp(r * DELTA)
    V_j = Y + h * S_j - M_PUT * put(tau_j, S_j)
    daily_loss[:, j - 1] = V_prev - V_j
    # re-hedge at T_j for the next period: new delta, cash absorbs the share
    h_new = M_PUT * put_delta(tau_j, S_j)
    Y_new = Y + (h - h_new) * S_j
    h, S_prev, V_prev = h_new, S_j, V_j

L10 = daily_loss.sum(axis=1)                    # = V_t - V_{t+10 Delta}
assert np.allclose(L10, V_t - V_prev)
var_10d = var_emp(L10, ALPHA)
print('10-day loss (re-hedged daily): mean %.3f   sd %.3f' % (L10.mean(), L1
print('10-day VaR_0.95 (time aggregation) = %.4f' % var_10d)
```

```
10-day loss (re-hedged daily): mean 0.111   sd 48.001
10-day VaR_0.95 (time aggregation) = 88.6158
```

In [5]:

```
# the same paths with the hedge frozen at t for all ten days (the Homework 2
S10 = S_path[:, -1]
V10_static = h0 * S10 + (0.0 - h0 * S0) * np.exp(r * J_DAYS * DELTA) - M_PUT
L10_static = V_t - V10_static
var_10d_static = var_emp(L10_static, ALPHA)

print('one-day VaR x sqrt(10)        %8.2f' % var_sqrt)
print('10-day VaR, re-hedged daily   %8.2f   (ratio to sqrt rule %.3f)' % (v
print('10-day VaR, hedge frozen      %8.2f   (ratio to sqrt rule %.3f)' % (v

print('\nshape of the loss distributions')
```

```
for name, L in [('one-day', L1), ('10-day re-hedged', L10), ('10-day frozen
    skew = ((L - L.mean())**3).mean() / L.std()**3
    print('%-20s mean %8.3f  sd %8.3f  skew %6.3f  (VaR - mean)/sd %.3f'
          % (name, L.mean(), L.std(ddof=1), skew, (var_emp(L, ALPHA) - L.mea
print('\nsum of the daily losses: mean %.3f   sd of daily losses %s'
      % (daily_loss.sum(axis=1).mean(), np.round(daily_loss.std(axis=0, ddof
print('sd of the 10-day sum / (sqrt(10) x sd of day-1 loss) = %.3f'
      % (L10.std(ddof=1) / (np.sqrt(J_DAYS) * daily_loss[:, 0].std(ddof=1)))
```

```
one-day VaR x sqrt(10)           94.24
10-day VaR, re-hedged daily      88.62   (ratio to sqrt rule 0.940)
10-day VaR, hedge frozen        306.41   (ratio to sqrt rule 3.252)

shape of the loss distributions
one-day              mean   -0.023  sd   14.862  skew  2.834  (VaR - mean)/s
d 2.007
10-day re-hedged     mean    0.111  sd   48.001  skew  0.893  (VaR - mean)/s
d 1.844
10-day frozen hedge  mean    0.849  sd  156.607  skew  3.022  (VaR - mean)/s
d 1.951

sum of the daily losses: mean 0.111   sd of daily losses [15.48 14.91 15.12
15.09 15.11 15.26 14.99 15.2  15.28 15.5 ]
sd of the 10-day sum / (sqrt(10) x sd of day-1 loss) = 0.981
```

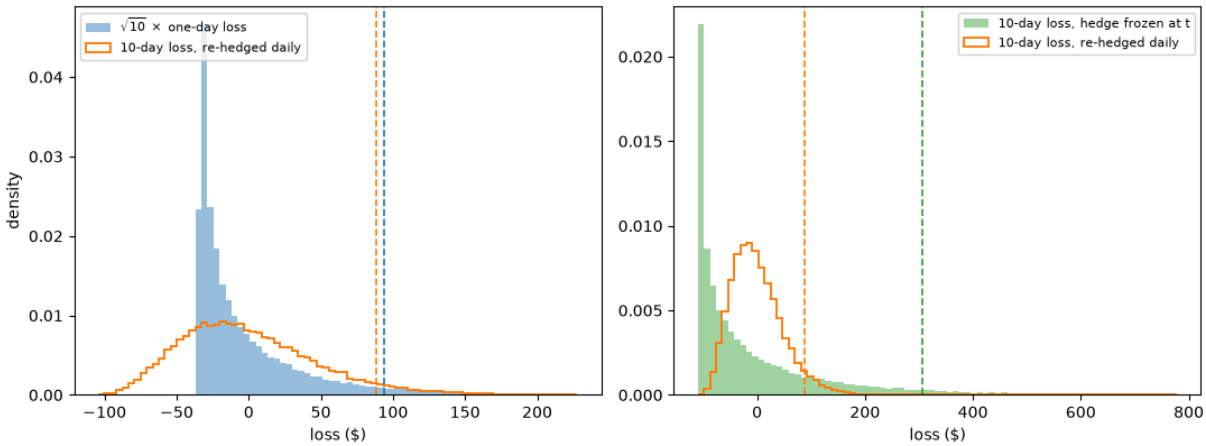
In [6]:

```
fig, ax = plt.subplots(1, 2, figsize=(11, 4.2))

top = max(np.quantile(L10, 0.995), np.quantile(np.sqrt(J_DAYS) * L1, 0.995))
bins = np.linspace(min(L10.min(), np.sqrt(J_DAYS) * L1.min()), top, 80)
ax[0].hist(np.sqrt(J_DAYS) * L1, bins=bins, density=True, alpha=0.45, label=
ax[0].hist(L10, bins=bins, density=True, histtype='step', lw=1.4, label='10-
ax[0].axvline(var_sqrt, color='C0', ls='--', lw=1.2)
ax[0].axvline(var_10d, color='C1', ls='--', lw=1.2)
ax[0].set_xlabel('loss ($)')
ax[0].set_ylabel('density')
ax[0].legend(fontsize=8)

bins2 = np.linspace(L10_static.min(), np.quantile(L10_static, 0.995), 80)
ax[1].hist(L10_static, bins=bins2, density=True, alpha=0.45, color='C2', lab
ax[1].hist(L10, bins=bins2, density=True, histtype='step', lw=1.4, color='C1
ax[1].axvline(var_10d_static, color='C2', ls='--', lw=1.2)
ax[1].axvline(var_10d, color='C1', ls='--', lw=1.2)
ax[1].set_xlabel('loss ($)')
ax[1].legend(fontsize=8)

fig.tight_layout()
plt.show()
```


√10 × one-day loss
10-day loss, re-hedged daily
density
loss ($)
10-day loss, hedge frozen at t
10-day loss, re-hedged daily
loss ($)